

Lecture: 13 Expansion of Functions

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Maclaurin's Theorem

Let $f(x)$ be a functions of x which can be expanded in powers of x and let the expansion be differentiable term by term any number of times.

Suppose, $f(x) = A_0 + A_1x + A_2x^2 + A_3x^3 + \dots$

Then by successive differentiation, we have

$$f'(x) = A_1 + 2A_2x + 3A_3x^2 + 4A_4x^3 + \dots,$$

$$f''(x) = 2.1A_2 + 3.2A_3x + 4.3A_4x^2 + \dots,$$

$$f'''(x) = 3.2.1A_3 + 4.3.2A_4x + \dots; \text{ etc.}$$

Putting $x = 0$ in each of these, we get

$$f(0) = A_0, f'(0) = A_1, f''(0) = 2!A_2, f'''(0) = 3!A_3, \dots$$

Hence,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

This is known as Maclaurin's Theorem.

Example. Expand $e^{\sin x}$ by Maclaurin's theorem.

Solution. Let

$$f(x) = e^{\sin x}, \text{ then } f(0) = 1,$$

$$\Rightarrow f'(x) = e^{\sin x} \cdot \cos x \Rightarrow f'(0) = 1$$

$$f''(x) = -e^{\sin x} \sin x + \cos x e^{\sin x} \cdot \cos x \Rightarrow f''(0) = 1$$

$$\Rightarrow f'''(x) = -e^{\sin x} \cos x - \sin x \cdot e^{\sin x} \cos x - 2 \cos x \sin x e^{\sin x} + \cos^2 x e^{\sin x} \cos x \Rightarrow f'''(0) = 0.$$

By Maclaurin's theorem, we have

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

$$\Rightarrow e^{\sin x} = 1 + x + \frac{x^2}{2} + \dots$$

Taylor's Theorem

Let $f(a+h)$ be a function of h which can be expanded in powers of h , and let the expansion be differentiable any number of times with respect to h . Suppose,

$$f(a+h) = A_0 + A_1h + A_2h^2 + A_3h^3 + \dots$$

By successive differentiation with respect to h , we have

$$f'(a+h) = A_1 + 2A_2h + 3A_3h^2 + 4A_4h^3 + \dots$$

$$f''(a+h) = 2.1A_2 + 3.2A_3h + 4.3A_4h^2 + \dots$$

$$f'''(a+h) = 3.2.1A_3 + 4.3.2A_4h + \dots \text{ etc.}$$

Putting $h = 0$ in each of these, we have

$$f(a) = A_0, f'(a) = A_1, f''(a) = 2!A_2, f'''(a) = 3!A_3, \dots$$

Hence

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \dots + \frac{h^n}{n!}f^{(n)}(a) + \dots$$

This is known as Taylor's Theorem. If we put $a = 0$ and $h = x$, we get the particular case known as Maclaurin's Theorem.

A more general form of Taylor's theorem is

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots + \frac{h^n}{n!}f^{(n)}(x) + \dots,$$

which is obtained by replacing a by x in above expression.

A more general form is obtained by replacing h by $(x-a)$ in the above expression. Thus,

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \dots,$$

which is an expansion in power of $(x-a)$.

Example. Expand $2x^3 + 7x^2 + x - 1$ in powers of $x - 2$.

Solution. Let $f(x) = 2x^3 + 7x^2 + x - 1$.

$$f(x) = 2x^3 + 7x^2 + x - 1 \Rightarrow f(2) = 2.2^3 + 7.2^2 + 2 - 1 = 45$$

$$f'(x) = 6x^2 + 14x + 1 \Rightarrow f'(2) = 6.2^2 + 14.2 + 1 = 53$$

$$f''(x) = 12x + 14 \Rightarrow f''(2) = 12.2 + 14 = 38$$

$$f'''(x) = 12 \Rightarrow f'''(2) = 12$$

Taylor's series is

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots$$

Here, $a = 2$

$$\begin{aligned} f(x) &= 45 + (x-2).53 + \frac{(x-2)^2}{2!}.38 + \frac{(x-2)^3}{3!}.12 \\ \implies f(x) &= 2x^3 + 7x^2 + x - 1 = 45 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3. \end{aligned}$$

Example. Expand $\sin x$ in powers of $(x - \pi/2)$.

Solution. As we know by Taylor's theorem

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots + \frac{(x-a)^n}{n!}f^n(a) + \dots$$

We have to expand $\sin x$ in power of $(x - \pi/2)$ hence $a = \pi/2$. Let

$$\begin{aligned} f(x) &= \sin x \Rightarrow f(\pi/2) = \sin(\pi/2) = 1, \\ f'(x) &= \cos x \Rightarrow f'(\pi/2) = \cos(\pi/2) = 0, \\ f''(x) &= -\sin x \Rightarrow f''(\pi/2) = -\sin(\pi/2) = -1, \\ f'''(x) &= -\cos x \Rightarrow f'''(\pi/2) = -\cos(\pi/2) = 0, \dots \end{aligned}$$

Hence,

$$\sin x = 1 - \frac{\left(x - \frac{\pi}{2}\right)^2}{2!} + \frac{\left(x - \frac{\pi}{2}\right)^4}{4!} + \dots$$

Expansion by Algebraic and Trigonometrical Methods

Sometimes it is more convenient to use the Binomial, Exponential or the Logarithmic theorems, or the expansion of $\sin x$ and $\cos x$. But care should be taken not to omit inadvertently any term of an order lower than, or the same as, that of the highest order term retained.

Example. Expand $(\tan x)^{1/3}$.

Solution. Expansion of $\tan x$ is

$$\tan x = \left(x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots \right)$$

So,

$$(\tan x)^{1/3} = \left(x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots \right)^{1/3}$$

$$\begin{aligned}
&= x^{1/3} \left[1 + \frac{x^2}{3} + \frac{2}{15}x^4 + \dots \right]^{1/3} \\
&= x^{1/3} \left[1 + \frac{1}{3} \left(\frac{x^2}{3} + \frac{2}{15}x^4 + \dots \right) + \frac{\frac{1}{3}(\frac{1}{3} - 1)}{2} \left(\frac{x^2}{3} + \frac{2}{15}x^4 + \dots \right)^2 + \dots \right] \\
&= x^{1/3} \left[1 + \frac{x^2}{9} + \frac{2}{45}x^4 + \dots - \frac{1}{81}x^4 - \dots \right] \\
&= x^{1/3} \left[1 + \frac{x^2}{9} + \frac{13x^4}{405} + \dots \right].
\end{aligned}$$

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.